FilingRevisionQA: Supplementary Documentation

This separate supplementary document accompanies the paper and is not included in the public resource core. The project repository has been created; completeness of the public core upload is being verified. The core uses scoped MIT and CC BY 4.0 licenses and names ZiBo Zhang as maintainer. Sections prefixed S are cited from the main paper. Separate upload of this PDF as a conference supplement has not been confirmed; the software-artifact field is not treated as permission for additional paper pages.

This supplementary document links the paired resource evaluation to its construction records, controlled experiments and reproduction procedures. For construction and human-review provenance, start with S11. S1–S4 present adaptation, control and external-task results; S5–S7 document interpretation boundaries, package verification and scoring revisions; S8–S10 provide paired-outcome displays and retained-output breakdowns. S12 summarizes related resources; S13 preserves the full evaluation settings and counts; S14 retains the fuller method and interpretation wording behind the condensed main text.

S1 Source binding and fixed adaptation

S1.1 Source binding changes overall and paired outcomes

Source binding improves overall development correctness while reducing success on the ten changed development states (Table S1). Rules extract period, unit, and scope. The question-only LLM parser uses the same ranking procedure and falls back to rules on invalid output.

Table S1. Source binding raises overall correctness while reducing changed-state correctness.

Development condition	Joint /144	Both /72	Changed /10
Frozen recovery	78	35	4
Rules + lexical fact ranking	95	40	4
LLM slots + same ranking	94	39	4
Rules + ranking + single-accession binding	102	48	2
LLM slots + ranking + single-accession binding	102	48	2
Rules + binding + source context	101	49	4
Source context with numbers masked (diagnostic)	103	47	5

Note. Conditions use the same Qwen reader and budget caps. Binding uses public relevance and acceptance-time tie-breaking. Evidence lengths differ. The final row is a post-hoc masking control; ten changed states represent five issuers.

The LLM parser changes joint correctness by -1 without binding and zero with binding, both adjusted p = 1.0. Its 49/72 mechanically valid proposals are a parser-acceptance count. Binding changes changed-state success from 4/10 to 2/10. The resource thus distinguishes a gain in overall answer compliance from a gain on the small changed subset.

Reference-isolated auditing identifies the transition behind binding's result. Of 122/144 mapped states, all mapped facts meet the cutoff, 108 survive filters, and top-five target presence rises from 87 to 93. Seven of ten gains repair citation consistency while preserving number and accession; two repair accession and one adds a target fact. All three losses already contain the correct fact. The remaining 22 states have unknown mapping status.

Adding context to the same five facts gives six gains and seven losses. Every loss equals reference/1000 despite the correct unit name and accession. Masking numbers partly reduces these scale errors while also removing dates and

shortening inputs. The contextual change therefore exposes a scale-sensitive failure mode, with the intervention's attribution limits retained in §6.1.

S1.2 Adaptation improves supported-interface development scores

Fixed adaptations raise overall scores on the supported training interface. Public rule-bound inputs were frozen before existing train references were mapped to citations. Of 582 training states, 281 satisfy the interface: 245 optimization and 36 loss-holdout states from 47/8 issuers. Four predeclared near-template cases and 297 without typed/exact support are omitted, retaining their exclusion status rather than creating abstention targets.

Each LoRA run [25] uses the final model after one epoch, rank 16, alpha 32, dropout 0.05, and learning rate 0.0001. Seeds 43/44 repeat seed42 with identical data and configuration. Evaluation retains all 144 development states and the exact unadapted inputs.

Table S2. Fixed adaptation seeds yield higher overall scores than their unadapted controls.

Reader	Base J	J: 42/43/44	Mean % ± SD	P: 42/43/44
Qwen	102	111/112/112	77.55 ± 0.40	54/55/55
Phi	90	106/108/106	74.07 ± 0.80	49/53/51

Note. J uses 144 states; P uses 72 pairs. SD is the sample standard deviation in percentage points. All fixed seeds are reported; statistical sampling units follow §5.1.

Changed-state successes are 5/6/5 for Qwen and 5/5/5 for Phi out of ten. Seed42 gains are 6.25/11.11 percentage points, with both Holm-adjusted sign-flip p-values at 0.127. A first-card-copy control reaches 101/144 joint, 49/72 paired, and 5/10 changed correctness, including 14 invalid outputs. This control makes copying behavior an explicit comparison for interpreting the adapted readers.

Exact-source support is assessable for 83/144 inputs, while 61 remain unknown. Qwen joint successes with a designated citation vary across seeds, 77/67/76, despite similar overall scores. Later context training obtains 115/115/116 for Qwen and 109/109/110 for Phi. Same-filing/concept/unit, different-period distractors yield 115/114/116 and 108/111/110 under the fixed 245/36 split and one-epoch configuration. Mean differences are -0.23/+0.23 points, both Holm p = 1.0; changed counts remain 6/10 and 5/10. These separately registered conditions add a negative result on the proposed period intervention.

S2 Coverage and identifier controls

Internal-holdout evaluation shows where supported-interface gains transfer. The inspected holdout includes 86 states from 14 training-partition issuers excluded from optimization: 36 supported loss-holdout states, 42 without exact binding, and eight raw fallbacks. Seed42 adapters change Qwen 39 to 41/86 and Phi 36 to 41/86. Each obtains 34/36, 6/42, and 1/8 across these strata. Qwen abstentions fall from 26 to zero while valid joint errors rise from 12 to 31. The two USD/share states retain their denominator positions and receive no joint credit after native-unit rescoring.

Citation identifiers also affect the outcome. Cycling IDs while preserving all other bytes and tokenizer lengths leaves Qwen base/adapter joint counts at 102/111. Phi changes from 90/106 to 71/110, with 21 new base-model accession/citation mismatches. Invalid outputs receive no equivariance credit. The control isolates this fixed identifier permutation, complementing the content-access comparisons.

S3 Output-format controls

Protocol validity and joint correctness respond differently to output constraints. We cross unadapted weights and a retained seed42 adapter with free greedy decoding, a JSON-envelope grammar, and the envelope plus a numeric-string constraint. The grammar permits null; reference values and target accessions remain offline. All six arms per family are generated in the same runtime with unchanged prompts, evidence order, and budgets.

Table S3. Numeric output constraints add joint successes on unchanged-answer states.

Model and weights	Free J	Envelope J	Numeric J	P free to numeric	Changed J free to numeric
Qwen unadapted	103	103	103	49 to 49	2 to 2
Qwen adapted	111	111	113	54 to 54	5 to 5
Phi unadapted	88	90	94	41 to 44	4 to 4
Phi adapted	109	109	110	52 to 53	6 to 6

Note. J uses 144 states; P uses 72 pairs; changed J uses ten states. All controls are newly generated in the same runtime, separate from the earlier Table S2 runtime. Supplementary Section S5.4 records software and cost.

All four predeclared numeric-minus-free comparisons have issuer-cluster Holm p = 1.00, with intervals including zero. Adapted Qwen gains 14 valid responses, from 130 to 144, and two joint successes; adapted Phi gains four valid responses, from 140 to 144, and one joint success. All nine additional joint successes across the comparisons occur in unchanged-reference states. Under identical numeric constraints, adapted-minus-base gaps remain 10/144 and 16/144.

These results separate output compatibility from the paired factual target: the tested constraints close validity gaps more readily than joint-answer gaps. Equivalence and the causal share of formatting remain unresolved under this design. Both constrained decoders increase latency.

S4 External FinQA

S4.1 Fixed retriever-generator replication

The external FinQA study [1] evaluates numerical programs using three complete retriever-generator seeds with the same scientific code, upstream revision, decoding hyperparameters, and 1,147 test identities. Within each seed, FQ2 and the FQ3 retrieval-replacement condition share the selected generator. FQ2/FQ3 are internal condition names. Retriever-produced evidence and selected weights vary across seeds, so the replication concerns the complete pipeline.

Table S4. FQ2 exceeds FQ3 in execution and program correctness across seeds.

Seed	FQ2 execution	FQ2 program	FQ3 execution	FQ3 program
42	699	670	516	499
43	697	667	512	489
44	724	696	530	510
Mean accuracy percent	61.61	59.08	45.28	43.53
Sample SD percentage points	1.31	1.39	0.82	0.92

Note. Every count uses 1,147 test questions. All 6,882 condition predictions were rescored by the original official evaluator; the repeated seeds reuse the same questions rather than create 3,441 independent observations.

FQ3 loses 16.33 percentage points of mean execution accuracy. Under this frozen replacement, FQ2 retains the stronger result. Seed43's authorized packaging-only continuation preserved one unterminated development program, its incorrect outcome, and the full 883-item denominator. Training, tokens, beam choice, and checkpoint choice were unchanged. The linked continuation and seed44 original run each complete 17 stages; the original failure remains in the backup provenance.

S4.2 Coder adaptation

The separate Qwen2.5-Coder study [26] uses official FinQA training examples, with 3,183 optimization and 487 internal-holdout examples, one epoch, and 883 development inputs. Same-input unadapted execution is 109/883. Initial LoRA seeds42/43/44 obtain 533/533/504, mean 59.27%, sample SD 1.90 points, with program counts 473/466/460. FinQANet's 572/883 is a different-architecture reference.

Separately registered lexical-, random-, and neural-retrieval-distractor variants obtain 562/557/550, 576/559/546, and 570/562/569; the latter mean is 567/883. Each configuration retains its own seed summary. A single best seed is not a stable architecture comparison. These experiments establish performance on the external numerical-program interface; the SEC paired task and its gate remain separate, and pretraining contamination remains unresolved.

S5 Audit and retrospective analyses

S5.1 Literature-reading coverage

Comparisons concern implemented task interfaces and measurement targets; performance numbers from unmatched tasks are not compared. Same-Number Citation Swaps supplies the explicit cited-role comparison discussed in main §2 [20]. The targeted source review retains inaccessible materials and search limits, rather than asserting exhaustive coverage.

AhaSignals' timing resource contains eight real SEC cases and 16 prompts. Its 500-scalar-pair study contains 17 differences and 483 unchanged controls. Retrospective document review covers the 17 differences and three controls. Thirteen differences are precision-compatible and four exceed the rounding interval; causes remain unresolved for all 17. The comparison uses inspected author resource documentation because the linked SSRN manuscript was inaccessible. The Bai and Chen comparison uses their released resource and supplement rather than an unavailable main article. These documented capabilities establish precedents for real SEC versions, unchanged financial controls, and source traceability; Table S13 describes implemented tasks without assuming earlier resources could not be extended.

S5.2 Human and source records

The author has confirmed completion of human review for the mechanically projected public wordings. This confirmation is retained as a batch-level attestation linked to the exact projected-ID set and wording hashes. It supplies no individual reviewer identity, review date or item-level decision file; these fields remain unfilled. A separate technical audit checks transformations, paired inputs and reference/source mappings without adding human judgments.

Machine-assisted construction and repairs were provisional and evaluation-ineligible before human review. All retained IDs descend from the initial inventory. AI screening routes record processing decisions rather than a human error-rate estimate or standardized human exclusion taxonomy. The adjudicator's access to both returns makes adjudication informed rather than independently blind. Author attestations describe reviewer behavior; the process itself was not directly observed. Blank human labels remain blank.

Raw reviewer agreement measures field strings, not semantic agreement or correctness. Unresolved categories remain outside resolved-class claims; the full event collection is not labeled confirmed restatements. Public reviewer records exclude identities and correspondence. Structural checks and exact source reconstruction verify parsing and localization under retained software, while semantic interpretation remains a separate review question.

S5.3 Unit profiles, failures, and secondary analyses

The main tables use resource-native-v1. Historical autonomous contrasts remain unchanged; native complete-source figures are retained-output sensitivity estimates, with historical corrected contrasts separately registered.

The historical profile supports USD and shares, leaving four test pure states without joint credit. Uniform rescoring covers 25,930 canonical QA outputs in 173 condition/cohort groups, including retained ordinary-test rounds. It

changes neither values, references, parser outcomes, nor generations. Non-QA choice/likelihood records and unmapped synthetic records are outside this audit. The two complete-source Qwen gains are the sole joint-score changes; original profiles and historical inferential statistics remain identifiable.

Invalidity remains part of the primary scoring denominator, and the first-failure sequence preserves citation inconsistency separately from later accession mismatch. A mismatched accession neither establishes factual falsehood nor universal supersession of unchanged original text. Secondary readable-number counts leave primary scores intact. Semantic explanations and error attributions remain provisional; unknown source mappings are not automatically errors. Cases are selected deterministically by ID hash within observed outcome categories to illustrate retained outcomes rather than estimate their frequency. Number masking also removes dates and shortens input, while source containers differ in composition and coverage; these controls describe their actual interventions.

S5.4 Experiment history, repairs, and compute

The registry preserves the original one-fact ordinary-test study and 840 outputs, representation searches, all adaptation families, and their failures. Legacy machine-reference experiments remain separate from the human-reviewed cohort. The compact-five supplementary comparison retains Qwen 90/210 and changes Phi 81/210 to 76/210, both adjusted $p = 1.0$. These histories are neither discarded nor pooled as independent replications.

The cross-model parser type-check repair reuses 31 outputs without changing inputs or acceptance rules. One generated output was not persisted, leaving a recorded audit gap; subsequent generations are journaled before parsing and all 1,770 logical records remain. A context experiment changed key order in 14 unchanged fallbacks; reconciliation reuses exact original outputs to produce 101/144 without new generation. Both versions remain separately archived. Engineering reconciliation and unit rescoring add no model replication.

Adaptation used an RTX PRO 6000 for Qwen and RTX 4090 for Phi, with reserved peaks of 37.76 to 42.31 and 13.32 to 15.48 GiB. The format factorial used two RTX 4090 hosts with PyTorch 2.9.0+cu128, Transformers 4.51.3, PEFT 0.15.2, and lm-format-enforcer 0.11.3. It retained 1,728 actual calls, zero cache reuse, and 105,705 generated tokens. Summed generation-loop time was 4,194.92 seconds, with loading, hashing, and archiving additional. Fixed execution order and load limit latency comparisons; the compute registry distinguishes stage wall time, model loops, missing measurements, and rental allocation.

S5.5 Reuse and release status

Offline reuse checks and pinned lxml/libxml2/libxslt reconstruction retain failures and successful cross-platform results. They establish the tested offline path; live SEC source availability is a separate check. Reviewed-wording lineage and evaluated inputs are preserved. Main Section 3.5 gives the repository, scoped licenses and maintenance contact. Authors and affiliations follow author-supplied metadata; funding details remain unprovided. Third-party licenses continue to apply, and the historical reproduction packages remain distinct from the public core.

S6 Historical package verification

The resource preview supports component replacement and source-level answer inspection. Its Python 3.10+ loader validates membership and hashes. Public loading exposes question, cutoff, clock protocol, and instruction; experimental runners add system instructions and retrieved evidence. Offline references and review provenance remain separate.

Resource cards, examples, and license inventories accompany 336 filings, 938 source URLs/hashes, and 1,555 XML records with contexts and units. Retrieval uses 70 development and 84 test filings. The project SEC client fetches official URLs and checks expected hashes; pinned parsing assets support exact historical extraction. A standalone extraction outside the checkout passes 42 tests and eight commands without third-party packages or network access, covering 936 references, 468 wording-lineage records, a 938-file source plan, and exact replay of 144 retained outputs with J102/P48.

The new-component helper seals new requests. Reconstructing historical prompts uses the retained historical assets; the helper is a separate entry point. API separation is not a secrecy guarantee, and excerpt entailment and semantic sufficiency require their own checks. Main Section 3.5 identifies the core's access point, scoped licenses and maintainer; the historical package exclusions remain documented here.

The publication core (v1.0.0, documentation revision r3) preserves the same data and scoring code with a loader, schema, resource card, official-source download plan and a synthetic no-evidence interface example. Its clean extraction passes 67 tests and seven commands without network or GPU access. It excludes the historical source-excerpt/prediction replay, base weights and environments; the earlier replay results above refer to the historical package. The project repository has been created; completeness of the public core upload is being verified. Scoped license grants and the maintainer are recorded in the core. Live SEC source restoration remains separate from repository download verification.

S7 Retained-output diagnostic repair and wording strata

S7.1 Versioned repair

The revised package makes paired comparability explicit. The earlier portable scorer could classify two protocol-valid outputs with an unrecognized unit as a false change, because failure to establish equality was treated as inequality. The revised diagnostic requires both units to be recognized; otherwise the pair is unassessable. Number, unit, accession and primary joint credit use the unchanged resource-native-v1 profile. Historical code and outputs are retained.

The frozen inventory covers 164 retained files: the 152-file historical inventory and the twelve accepted format-control arms. Across 27,658 canonical state/condition records, the revision reproduces 25,930 historical state scores and 12,965 historical pair diagnostics exactly. Relative to the earlier portable implementation, 239 repeated pair/condition records in 128 groups change from false-change to unassessable. These are not 239 independent questions. Historical main-table scores and the published-in-draft false-change counts remain unchanged because the earlier manuscript audit already used the comparability guard. Synthetic and non-QA identifiers remain separately recorded rather than mapped into the canonical cohort. This is a uniform retained-output audit with no answer generation or model selection.

S7.2 Wording-stratified description

Table S5 partitions the evidence-recovery outputs by existing wording lineage and reference change. It describes the same exposed ordinary test; wording category also varies with question and issuer composition. It neither establishes semantic equivalence nor isolates a wording effect.

Table S5. Wording lineage separates the composition of retained recovery outcomes.

Wording	Reference change	States / pairs / issuers	Qwen J / P	Phi J / P
Cutoff projection	Unchanged	40 / 20 / 6	10 / 4	10 / 4
Cutoff projection	Changed	2 / 1 / 1	0 / 0	0 / 0
Prefix projection	Unchanged	64 / 32 / 9	39 / 17	33 / 10
Prefix projection	Changed	6 / 3 / 3	2 / 0	2 / 0
Verbatim	Unchanged	84 / 42 / 14	36 / 15	35 / 14
Verbatim	Changed	14 / 7 / 6	3 / 1	1 / 0

Note. J uses the state denominator and P the pair denominator in each row. Issuers can recur across strata and are not additive. Full-condition issuer-bootstrap intervals and row counts accompany the machine-readable analysis. A one-issuer stratum or all-zero empirical resamples can yield a degenerate interval; that interval is not an independent population-precision estimate. These strata support inspection of wording provenance, with the completed-review confirmation recorded in S5.2; they are not used for retrospective exclusion or re-selection.

S7.3 Structural coverage profile

Table S6 reports only explicit retained fields. Filing acceptance year differs from reporting year, and source modality describes record structure rather than evidence sufficiency.

Table S6. Explicit metadata describes source-year, unit and locator coverage.

Field	Value	Count	Unit
Filing acceptance year	2019	1	filings
Filing acceptance year	2020	7	filings
Filing acceptance year	2021	109	filings
Filing acceptance year	2022	65	filings
Filing acceptance year	2023	58	filings
Filing acceptance year	2024	64	filings
Filing acceptance year	2025	32	filings
Reference unit	iso4217:CAD	8	states
Reference unit	iso4217:USD	848	states
Reference unit	iso4217:USD per xbrli:shares	6	states
Reference unit	xbrli:pure	8	states
Reference unit	xbrli:shares	66	states
Locator modality	html	76	states
Locator modality	html+xml	860	states

Note. No form field is present in the public filing metadata used for this audit; form counts are not inferred from filenames. A semantic question-type taxonomy is likewise unreported. These are metadata-coverage boundaries, not zero-count categories. The original resource, references and split membership are unchanged.

S7.4 Reproduction entry points

The revised candidate retains the historical scorer and adds paired_diagnostics_v2.py. QUICKSTART_NEW_COMPONENT.md documents public loading, evidence validation, message sealing, reader outputs and offline scoring. build_component_requests.py constructs a new component's interface without reading reference answers. Historical experimental inputs remain separately preserved. The audit freeze, per-state and per-pair results, exclusions, difference ledger, and scripts accompany this revision; replay consumes retained outputs rather than calling a model.

S8 Visual comparison of paired outcomes

Figure S1 renders Table S14 without changing its counts. Each bar retains all 105 pairs; both-state success, one-sided success and shared failure sum to the same denominator.

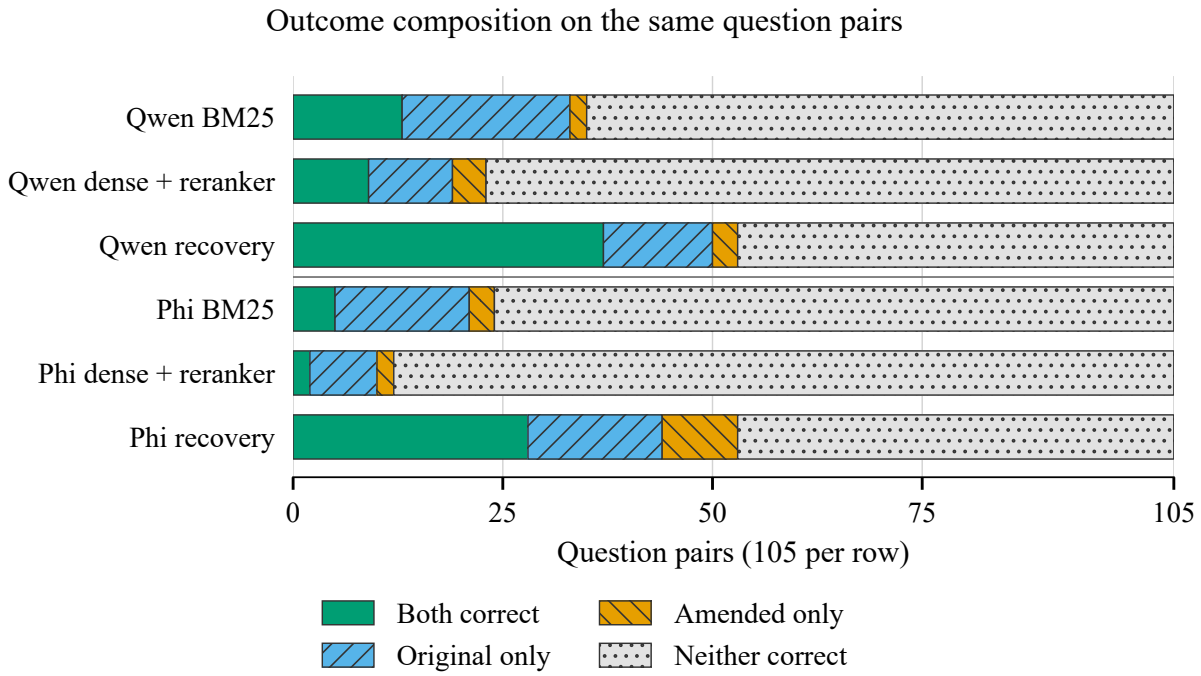


Figure S1. Paired outcomes show directional asymmetry across the six autonomous conditions. Counts are the retained outcomes of 105 ordinary-test pairs per condition; repeated conditions are not additional independent questions.

S9 Numerical matches in retained parsed fields

The retained-field diagnostic separates readable answer matches from the full response contract. It compares the answer string and canonical unit in preserved parsed JSON, independently of the recorded protocol status. Raw outputs, Markdown and prose are not reparsed or repaired. Table S7 uses resource-native-v1 and retains every scheduled state.

Table S7. Retained answer fields reveal numerical matches outside the primary protocol gate.

Reader / condition	Readable N/210	Readable NU/210
Qwen · Cutoff BM25	64	60
Qwen · Cutoff dense + reranker	42	39
Qwen · Evidence recovery	95	91
Qwen · Correct document + BM25 (R)	60	56
Qwen · Complete-source containers (R)	158	156
Phi · Cutoff BM25	49	44
Phi · Cutoff dense + reranker	30	27
Phi · Evidence recovery	93	92
Phi · Correct document + BM25 (R)	51	46
Phi · Complete-source containers (R)	119	117

Note. Each row retains all 210 states. R denotes the same reference-assisted conditions as main Table 4. Recognized-unit matching uses the native profile; unavailable parsed fields receive no match. These secondary counts leave primary scores unchanged and establish field agreement, while evidence support remains a separate target. Legacy and native diagnostics remain separately recorded; the Qwen complete-source unit-profile difference is documented in §5.1 and S5.3.

S10 Evidence composition and retrospective analysis

The following breakdowns describe cohort lineage, gain heterogeneity, candidate pools and scoring-stage outcomes.

S10.1 Version lineage and clock construction

The historical 400-pair review core and the current 468-pair cohort are different selections. Matching their retained source-question IDs yields 208 shared IDs, 192 unique to the old core and 260 unique to the current cohort. The current set is therefore not a simple expansion of that core. Table 1 instead follows the 890-to-485-to-468 construction route; the ID-level crosswalk is supplied with the revision evidence. Selection labels unavailable at earlier stages are not backfilled from the final changed-answer split.

All 936 state cutoffs equal the Accepted timestamp of their designated filing, and every pair has an earlier original cutoff. The public clock protocol is SEC_acceptance_proxy_not_verified_publication. This construction, issuer routing and sparse eligible filing pool supply strong task constraints; a source match is compliance with this declared task, not a measurement of unconstrained revision discovery.

S10.2 Retrospective gain heterogeneity

We resample 29 issuer clusters 20,000 times with replacement using seed20261029. Each draw carries both methods, both states and both models together, and retains every scheduled failure. Gains are recovery minus BM25; the contrast subtracts the unchanged-state gain from the changed-state gain. The changed stratum has 22 states/11 pairs from ten issuers; the unchanged stratum has 188 states/94 pairs from 29 issuers. All draws contain both strata.

Table S8. Recovery-gain contrasts describe different changed and unchanged outcomes.

Reader	Changed gain, pp [95% interval]	Unchanged gain, pp [95% interval]	Difference, pp [95% interval]
Qwen	-9.09 [-35.72, 14.29]	23.40 [9.20, 37.50]	-32.50 [-61.06, -2.24]
Phi	0.00 [-14.29, 13.64]	27.66 [18.27, 37.50]	-27.66 [-45.95, -9.22]

Note. Intervals are pointwise type-7 percentile 95% intervals, not familywise-adjusted confirmation. The analysis is retrospective on an exposed ordinary test and provides no causal difference-in-differences identification or new p-value. Changed-stratum gain intervals include zero for both readers; the contrast intervals describe heterogeneity on this sample. Model-to-model interaction differences were not tested.

Qwen corrects/regresses 1/3 changed states and 61/17 unchanged states; Phi corrects/regresses 1/1 and 61/9. Positive net gains from unchanged states therefore coexist with individual corrections on changed states. All draws, issuer sufficient statistics, state outcomes and input hashes accompany the computation freeze.

S10.3 Experimental candidate collections

Table S9. Recorded candidate pools distinguish documents, facts and natural passages.

Measurement	Development	Ordinary test
Scheduled states	144	210
Corpus filings	70	84
Fact index records	85,125	68,348
Fact-route filings: median [range]	2 [1, 3]	2 [0, 3]
Fact candidates: median [range]	1,802 [332, 6,963]	1,363 [0, 8,110]
Passage candidates: median [range]	2,082.5 [539, 7,564]	1,820 [0, 5,316]
Empty passage pools	0	10

Note. These are the retained experimental corpora, not all SEC filings or the full resource source registry. Fact routes were reconstructed from frozen public helpers; passage counts were linked to the exact original BM25/reranker top-20 records. Empty states remain scheduled failures, not newly assigned unanswerable labels. No future source enters the recorded routes. The few-document setting and specified cutoff contract limit document-discovery generalization.

S10.4 Complete-source error composition

Table S10. Complete-source errors differ across protocol and numerical layers.

Reader / split	States	Protocol / precondition	Valid N wrong	N right U wrong	NU right A wrong	J
Qwen dev	144	23	54	0	0	67
Qwen test	210	22	30	2	0	156
Phi dev	144	48	60	1	0	35
Phi test	210	56	38	2	0	114

Note. Counts are mutually exclusive under the retained resource-native-v1 scoring order and include all scheduled states. They describe reference-assisted source-container access; different split composition and context budgets prevent a causal attribution of the score gap.

S10.5 Recorded protocol and precondition failures

Table S11. Seven recorded statuses separate protocol and precondition failures.

Recorded status	Records
Abstained	209
Invalid numeric output	140
Citation/accession mismatch	132
No supplied evidence	60
Invalid JSON	55
Output length limit	15
Invalid schema	12
Total	**623**

Note. Six autonomous conditions reuse the same 210 test states, producing 1,260 condition-state records. These mutually exclusive retained statuses identify the first pipeline outcome. The 132 citation/accession mismatches are included in the 623 failures; the ten later designated-accession errors in Table S16 have already passed protocol, numerical and unit checks. This extraction reuses existing scores and does not classify all potential semantic defects.

S11 Construction and review records

S11.1 Resource record layers

Table S12. Linked record layers connect paired questions to their sources.

Record layer	Count	Role
Question pairs	468	Same question at two filing states
States	936	State-specific answer and source target
Issuers	144	Disjoint split membership
Filings	336	Identified source documents
Source URL/hash records	938	Exact source restoration plan
XML records	1,555	Context, unit and source lineage

Note. Counts refer to record layers in the retained resource snapshot. Source URL/hash records and XML records are different units. The split and changed-pair counts remain in Table 2.

S11.2 Review, wording and corrections

Source-grounded construction includes 67 stem repairs before independent review. The human return accepts 464 pairs and four after correction, preserving the submitted cohort's numerical answers, units, and adjudicated stems. Period annotations and ten locator repairs across four questions have separate records. Supplementary Section S5.2 distinguishes construction decisions, human judgments, and technical repairs.

Independent returns and adjudication records document the review process. The adjudicator inspected both returns. Author attestations confirm reviewer A's reading of both versions for every item, reviewer B's completion date, and human adjudication; original declarations and later attestations remain separately traceable.

The independent returns agree on question text, IDs, and both accessions for all 468 pairs and 144 events. Controlled-field agreement is 449/468 for question verdicts, 466/468 for period type, 144/144 for event verdicts,

and 130/144 for event category. The 19 question-verdict disagreements concern evidence-insufficient versus accepted judgments. Raw answer-cell agreement is 397/468 for originals and 396/468 for amendments. These are unnormalized string comparisons, sensitive to scale and formatting. We report raw agreement counts; the chance-corrected agreement coefficient κ was not computed.

The public question set preserves 240 reviewed wordings verbatim. A further 228 wordings use fixed mechanical projections: 167 replace an introductory filing phrase and 61 specify cutoff-based applicability. The author has confirmed completion of human review for these projected wordings. Original and public wordings, transformation categories and the confirmation remain traceable, and all reported predictions retain their original evaluated inputs. Technical checks verify the preserved text, paired-state wording and reference mappings. Four of eleven changed test pairs use projected wording. Supplementary Section S5.2 records the review evidence.

Eleven events containing 45 pairs retain unresolved categories. They participate in numerical QA; resolved-class analyses use the remaining 133 events and 423 pairs. Private reviewer identities and correspondence are excluded from the public fields.

S11.3 Source checks

Source hashes and locators make each transformation inspectable. Arelle checks cover 266 XML files, 1,555 locators, and 860 state records; 76 state records rely on HTML evidence.

Correction records preserve old and new fields, rationale, source evidence, reviewer provenance, and affected artifacts. Accepted human judgments, provisional machine analyses, unresolved categories, and technical repairs have separate record types. This structure supports both reconstruction of the evaluated resource and inspection of later corrections.

S12 Related-resource comparison

Table S13. Tasks and evaluation targets of related resources.

Resource / study	Comparison object	Documented evaluation or analysis
FinRank [11]	Question and supporting/distractor passages	Retrieval, reranking and hard-negative discrimination
AhaSignals timing benchmark [7]	Financial quantity at decision times	As-of answers and controlling accessions
AhaSignals scalar study and review [8, 9]	Historical scalar pairs	Value differences and selected context/precision review
VersionQA / FiscalQA Pro [13, 12]	Versioned technical / tax-law documents	Version-related answering and temporal grounding
AuditFraudBench [10]	Original/restated filings and enforcement material	Attribution, narrative and fraud-pattern tasks
FilingRevisionQA	Same question at original and amended filing states	Numerical, unit, designated-source and paired outcomes

Note. Rows follow the cited task descriptions; reading coverage and access limits are in S5.1. The table does not enumerate every capability of each resource.

S13 Detailed evaluation record

The following settings, counts and qualifications are retained from the prior draft. Main §5 presents the focused comparison; table numbers follow the new mapping. Historical profiles and experiment identities remain unchanged.

S13.1 Evaluation protocol and paired measures

The protocol evaluates the same question at the original and amended filing cutoffs. Each response is scored against its state-specific reference, and the two outcomes are then paired to measure success across both states.

State-level scores apply protocol validity V before assessing the answer. V includes citation/accession consistency with the supplied evidence; citation inconsistency is rejected at this gate before the state-designated accession is evaluated. Let N denote exact finite-decimal equality with the reference, U canonical-unit equality, and A equality with the state-designated accession. The primary scores average $V \times N$ for numerical correctness, $V \times N \times U$ for numerical-and-unit correctness, and $J = V \times N \times U \times A$ for joint correctness over all scheduled states. Thus numerical credit requires a protocol-qualified response, while the denominator includes all scheduled states.

Invalid outputs receive zero primary credit even when a readable number is present. Format/number invalidity, citation inconsistency, abstention, missing evidence, and length failures are reported separately. Secondary numerical diagnostics use retained parsed JSON fields, leaving Markdown and prose numbers unextracted and primary scores unchanged. Accession equality measures the designated-source requirement; factual truth, citation entailment, and the scope of an amendment require separate evidence.

The primary resource-native-v1 profile supports USD, shares, CAD, USD/share, and pure. A uniform retained-output audit compares it with the historical USD/shares profile. All autonomous main scores remain unchanged; the Qwen complete-source row changes from 154/210 to 156/210 and from 74/105 to 75/105 pairs. Supplementary Section S5.3 records profile coverage and the complete rescore procedure.

Pair-level scoring records how success is distributed across the two filing states. Both-state correctness averages $P = J(\text{original}) \times J(\text{amended})$ over question pairs. The mutually exclusive outcomes are both states correct, original only, amendment only, and neither. References determine changed and unchanged strata by number and unit.

Within unchanged pairs, a false change occurs when both predictions are valid, their units are recognized by the fixed unit profile, and their numbers or units differ. We report this count against both the full unchanged-pair denominator and that assessable subset. Within changed pairs, a stale amendment answer requires a protocol-valid prediction and a recognized unit; its number and unit must match the original reference and differ from the amended reference. Invalid responses, unknown units, missing outputs, and abstentions retain their own statuses. These rules distinguish answer stability from an absent answer, and original-reference reuse from an arbitrary repeated wrong value.

S13.2 Controlled evidence access and localization measures

The reference-assisted conditions supply different amounts of localization information. **Correct-document BM25** receives the reference accession and selects evidence within that document. **Complete-source containers** use reference locators to include a complete table or paragraph with headings and context. The container builder preserves rows and columns and never copies the answer field. These conditions measure performance under additional evidence access and are reported separately from autonomous answering. Within the five-item budget, containers cover 134/144 development and 200/210 test states; the other ten states in each partition remain failures. The cross-model diagnostic independently reconstructs 438 source containers from 106 hash-verified HTML files, covering 334 of 354 development/test states within the evidence budget.

Retrieval scores evaluate localization of the recorded supporting anchors. Reference locators define designated-source and known-target localization tasks, with neither pooled relevance judgments nor judged negative passages. We report known-target Hit@k and reciprocal rank on an explicitly defined candidate pool; unannotated support remains outside those target counts. A hit may also be a same-source alias matched by accession, value, unit, period and URL/hash/XPath consistency. These measurements assess known-target fact/source matching; semantic relevance and evidence sufficiency require additional judgments.

S13.3 Experimental scope and reader settings

The development/test experimental corpora contain 70/84 filings. Public issuer and cutoff checks, plus question-explicit form routing for structured facts, leave a median of two routed filings per state, with ranges of 1–3 and 0–3. Natural-passage baselines search median pools of 2,082.5 and 1,820 passages; structured facts form a different representation. Ten test states have empty routed pools and retain their denominator positions. Supplementary Table S9 gives the separate fact and passage counts for this few-document setting.

Pinned Qwen2.5-7B-Instruct and Phi-3.5-mini-instruct readers share evidence and semantic instructions [23, 24]. Each uses its native chat template, BF16 greedy decoding, batch size one, at most five evidence items, and an 8,192-token total budget with 512 generation tokens. Exact model revisions and input hashes accompany each run.

The experiment registry separates ordinary-test analyses, development ablations, reference-assisted diagnostics, and external FinQA studies. Within the cross-model round, 2,478 logical records comprise 2,291 fresh calls, 137 exact-message cache hits, and 50 precondition failures; these are combined with 1,062 accepted Qwen records. The resulting 3,540 model/condition/state records reuse the same question cohorts. Cache reuse is restricted to the same model and protocol, and every registered model/condition/state retains its scheduled outcome. Supplementary Section S5.4 records historical searches, repairs, and compute accounting.

S13.4 Statistical analysis

Issuer-level paired analysis preserves dependence among questions and filing states. The cross-model protocol fixes 10,000 issuer bootstraps and 100,000 paired issuer sign flips with seed 20261026. Six prespecified comparisons use Holm correction: Phi recovery against each retriever, and each reader's correct-document versus BM25 and complete-source versus correct-document contrast.

Seed SD reports training variability in percentage points on the question sample fixed across conditions and seeds. Period-distractor comparisons first compute candidate-minus-control outcome differences under matched training seeds, average those differences within each state, and then resample issuers and apply Holm2.

Earlier one-fact, parser, representation, and seed42 adaptation families retain separate corrections. Holdout/identifier intervals use 10,000 issuer bootstraps with seed 20261027. The uniform unit audit uses the same bootstrap count and seed for retrospective marginal intervals.

Retained outputs support a retrospective analysis by wording-transformation category. Every existing transformation category and changed/unchanged group is retained. Issuer-bootstrap intervals describe observed groups; single-issuer and boundary-degenerate resamples provide no independent estimate of population uncertainty. These analyses select no model, prompt or subset for a new test.

The retrospective gain-heterogeneity analysis uses 20,000 joint issuer resamples, seed20261029, retaining both methods, both states and both models. It reports every prespecified component and pointwise percentile interval; computation was fixed after the original result directions were known (Supplementary Section S10.2).

Layered and paired evaluation describe answer components, outcomes across filing states, and performance on changed-answer states. Evidence-access comparisons, development adaptations, source-linked cases and output controls then examine these outcomes in their respective settings.

S13.5 Layered scores separate answer outcomes

In the autonomous comparisons, recovery improves state-level joint and both-state correctness for both readers. Table 4 follows each condition from protocol validity through numerical and unit matches to designated-source and both-state correctness. This is a single frozen protocol on an exposed ordinary test, with the Qwen comparison reusing retained outputs.

Note. V, VN, VNU and J use all 210 scheduled states; P uses all 105 pairs. VN and VNU are protocol-qualified numerical and numerical-plus-unit matches. V includes citation/accession consistency, so these columns retain that source check. R marks reference-assisted access. Inputs, rules and resource-native-v1 are fixed; historical profile differences are in §5.1.

Recovery obtains 37/105 complete pairs for Qwen and 28/105 for Phi. In state-level joint correctness, Phi gains 24.76 percentage points over BM25, with paired issuer-cluster 95% interval [16.51, 33.66] and Holm p = 0.000080, and 31.90 points over dense retrieval, interval [22.82, 41.36], p = 0.000060. The next comparison asks which pairs contribute these gains.

S13.6 Paired evaluation locates the gains

Positive net recovery gains arise from unchanged-answer states. Tables S14 and S15 locate the change in outcomes rather than treating the aggregate score as a measure of revision response.

Table S14. Paired outcomes expose directional asymmetry and shared failures.

Autonomous system	Both	Original only	Amendment only	Neither
Qwen BM25	13	20	2	70
Qwen dense + reranker	9	10	4	82
Qwen recovery	37	13	3	52
Phi BM25	5	16	3	81
Phi dense + reranker	2	8	2	93
Phi recovery	28	16	9	52

Note. Each row partitions all 105 pairs into mutually exclusive outcomes. Historical and native scoring profiles give the same autonomous counts.

Table S15. Positive net recovery gains arise from unchanged-answer states.

System	Changed J /22	Unchanged J /188	Changed both /11
Qwen BM25	7	41	2
Qwen recovery	5	85	1
Phi BM25	3	26	0
Phi recovery	3	78	0

Note. The 11 changed pairs involve ten issuers; the 94 unchanged pairs involve 29 issuers, with overlap between strata. Paired states remain grouped.

Qwen's gain of 42 correct states decomposes into +44 unchanged and -2 changed states. Phi's +52 is entirely unchanged. Recovery produces false changes on 9/94 Qwen and 20/94 Phi unchanged pairs. Among unchanged-answer pairs, Qwen has 57 pairs with two valid outputs, of which 56 have comparable units; both counts are 62 for Phi. The assessable false-change rates are therefore 9/56 and 20/62. Original-reference reuse is 0/11 and 1/11. These counts describe different failure directions within the same paired task.

A retrospective issuer-paired analysis gives changed-minus-unchanged recovery-gain contrasts of -32.50 and -27.66 percentage points for Qwen and Phi. Supplementary Section S10.2 reports the pointwise intervals; the retained computation covers all 22 changed states.

The six autonomous conditions retain the same strict ordering under state and paired scores. Pairing contributes error structure rather than a ranking reversal: it reveals original-only success, amendment-only success, and joint failure. For example, under the protocol qualification rule, Qwen BM25 produces 55 numerical, 51 numerical-plus-unit, and 48 joint matches but only 13 complete pairs. Three otherwise unit-correct responses cite a different accession.

Protocol validity separates readable numerical matches from those counted by the primary scores. Citation/accession inconsistency and later accession mismatch occupy different scoring layers. Retained parsed

JSON contains 64/210 and 49/210 readable numerical matches for Qwen/Phi BM25, compared with primary 55/210 and 32/210. Recovery yields 95/210 and 93/210 readable matches versus primary 94/210 and 82/210. Table S16 separates the successive scoring outcomes under the validity rule in §5.1. Supplementary Tables S7 and S11 provide parsed-field numerical diagnostics and the seven recorded protocol/precondition failure statuses.

Table S16. The scoring sequence separates protocol, numerical, unit, and accession outcomes.

First-failure partition	Qwen recovery	Phi recovery	All six autonomous conditions
Protocol/precondition failure	75	43	623
Valid, N wrong	41	85	315
N right, U wrong	4	1	18
NU right, A wrong	0	0	10
Joint correct	90	81	294
Total scheduled	210	210	1260

Note. Outcomes are mutually exclusive in scoring order. The 1,260 condition-state records reuse 210 states. Citation/accession inconsistency accounts for 132 of 623 first-layer failures, in addition to the ten later accession failures.

S13.7 Evidence access separates evaluation conditions

The retrieval audit measures known-target coverage on development; the reference-assisted comparisons examine answering with supplied document or source context.

The retained same-representation retrieval audit ranks the original fact text with BM25 and MPNet over all eligible development candidates (Table S17). Exact annotated support maps to the index for 110/144 states; all 34 unmatched states remain in the denominator. This diagnostic uses the earlier fact-text representation, separate from the later cross-encoder passages.

Table S17. Known-target coverage separates lexical and dense retrieval on shared inputs.

Retriever	Target Hit@5 /144	Target Hit@50 /144	MRR
BM25	31	73	0.14981
MPNet dense	17	44	0.07726

Note. All 144 development states and the same fact-text pool are retained. The historical dense inputs include 215 facts truncated at 384 tokens. MRR uses the first matched target in the full routed pool; unmatched states receive zero. Existing issuer-bootstrap intervals for dense-minus-BM25 Hit@5 and Hit@50 are [-24.66, 4.55] and [-33.52, -5.07] percentage points. These are descriptive target-coverage intervals; truncation was not isolated as a causal factor.

Reference-assisted access changes performance at both document and source-container levels. Correct-document BM25 raises joint correctness from 48 to 56 for Qwen and 29 to 46 for Phi; adjusted $p = 0.1709$ and 0.0655 under the six-comparison correction. Native complete-source accuracy reaches 74.29% for Qwen, with issuer-cluster 95% interval [60.53, 86.57], and 54.29% for Phi, interval [42.08, 66.67]. Scoring-profile sensitivity and the separately registered historical contrasts are recorded in Supplementary Section S5.3.

Even with complete-source access, 32 of 188 valid Qwen responses and 40 of 154 valid Phi responses fail the native number/unit requirement among the full 210 states. The source condition therefore gives a concrete set of residual errors to inspect after reference-assisted localization.

The development pattern differs: complete-source Qwen/Phi scores of 67/144 and 35/144 are below recovery's 78/144 and 71/144. Restricting to the 134 common-coverage states gives 67 versus 74 and 35 versus 67. Ten states per split exceed the five-container budget; development also includes 38 mixed paragraph/table states compared

with 12 in test. These differences motivate examining evidence composition and reading behavior together. Container completeness and causal interpretation are addressed in §6.1.

The split difference also appears among valid outputs: complete-source Qwen has 54 numerical mismatches on development versus 30 on ordinary test; Phi has 60 versus 38. Supplementary Table S10 preserves the complete mutually exclusive failure accounting. These distributions motivate source- and question-level inspection rather than identify the cause of the difference.

S13.8 Development demonstrations separate overall and changed-state gains

The development source-binding comparison shows higher overall joint correctness and lower changed-state correctness. On 144 Qwen development states, rules plus ranking obtain 95 joint matches and single-accession binding obtains 102; changed-state correctness falls from 4/10 to 2/10. The LLM parser adds -1 and zero joint matches under the corresponding controls. Supplementary Section S1.1 preserves Table S1, all seven conditions and their source-level transitions.

Fixed one-epoch LoRA adaptations [25] use 245 supported training states and 36 internal loss-holdout states. Across seeds42/43/44, Qwen obtains 111/112/112 joint matches and Phi 106/108/106, compared with respective unadapted controls of 102 and 90 on the same 144 development states. Qwen's baseline with 102 joint matches is the rule-ranking-and-binding condition above. Supplementary Section S1.2 gives all paired results, sample SDs, copying controls and training-support exclusions; the development scope is specified in §6.1.

S13.9 Paired cases connect predictions to source cells

Traceable cases show how answer and source errors differ within a pair.

Figure 3 uses Q0691 under evidence recovery: Qwen returns both reference answers, while Phi cites the amended accession with the original value. In Q0296, both Turtle Beach covers report USD 197,795,766; Qwen BM25 gives that number but cites the original accession for the amended state. The examples retain official URLs, cutoffs, source hashes, context, and predictions. Source byte hashes and table locators are verified; the figure is a transcription rather than a SEC webpage screenshot. Cases were selected by fixed ID hashes within observed outcome categories (Supplementary Section S5.3).

S13.10 Output constraints separate response validity from joint correctness

On the development cohort, adapted Qwen gains 14 valid responses, from 130 to 144, and two joint successes under numeric constraints; adapted Phi gains four valid responses, from 140 to 144, and one joint success.

The six-arm development control crosses unadapted and seed42-adapted weights with free, JSON-envelope and numeric-string-constrained decoding (Table S3). Each family uses the same runtime, prompts, evidence and budgets across its six arms; references remain offline. These contemporaneous controls have a separate runtime from the earlier adaptations in Supplementary Table S2.

Note. J uses 144 states; P uses 72 pairs; changed J uses ten states. All controls are newly generated in the same runtime, separate from the earlier Table S2 runtime. Supplementary Section S5.4 records software and cost.

All four numeric-minus-free issuer-cluster Holm p-values are 1.00, with intervals including zero. Changed-state counts remain unchanged in all four rows. Supplementary Section S3 reports the decoder design and cost; §6.1 gives the limits on equivalence and causal interpretation.

S14 Detailed method and interpretation record

This section retains the fuller wording behind the concise main text. The main-paper table and figure identifiers refer to unchanged assets; these records add no experiments or claims.

S14.1 Reuse interface

The reuse interface connects user-supplied retrievers and readers to state and paired scoring. Users load the sealed snapshot and verify its hashes with the Python 3.10+ loader. The source client uses official URL/hash records to restore source files. Users construct eligible evidence, retain exact messages and raw outputs, and score the outputs offline against references.

Table 3. The reuse interface separates model inputs from offline evaluation.

Note. The helper seals requests for a new component. Exact historical input reconstruction additionally requires pinned extraction and retrieval assets; Supplementary Section S6 documents this distinction and the interface boundaries.

Paired diagnostics report change and stability under the resource-native-v1 unit profile. Unknown units remain unassessable for paired change/stability. For false-change rates among unchanged reference pairs, the full and assessable pair denominators are reported separately. The historical scorer and raw outputs remain intact. Retained-output parity and the exact diagnostic differences are documented in Supplementary Section S7.

The core is the v1.0.0 resource package described in Section 3.5, with scoped MIT/CC BY 4.0 grants and a named maintenance contact. Protected-gate data, private workbooks, model binaries and caches are excluded. The source snapshot preserves its historical status fields; current package authorization and license notices are recorded separately.

The proposed maintenance procedure preserves immutable correction tickets with prior answers, accessions, hashes, locators, and affected evaluations; semantic changes require human adjudication.

S14.2 Answering methods

Autonomous answering proceeds through filing routing, evidence selection and reader generation, applying the same frozen method and configuration independently to the evidence available at each cutoff. Figure 1 separates this procedure from offline evaluation.

Autonomous inputs and eligible filings

The task is to select the latest applicable filing for the requested financial fact under the query cutoff. Its fixed public instruction is: “Select the latest applicable filing for this financial fact under the stated acceptance-clock cutoff; the target accession is not supplied.” A system returns a finite numerical value, canonical unit, supporting accession, and citation under a fixed JSON schema, or abstains.

Autonomous retrieval uses the public question, issuer metadata, cutoff and eligible corpus. The public request contains question, cutoff, clock_protocol and instruction; issuer membership routes retrieval outside that request, and retrieved evidence is supplied separately to the reader. The two states differ in cutoff and eligible evidence. Public inputs omit collection dates. Reference records and gold locator/target roles are confined to offline evaluation and explicitly separated diagnostics. Filing evidence contains observable values and accessions, but the autonomous input does not identify an accession as the offline target.

Each pair fixes two availability states: c0 is the original target filing's SEC Accepted timestamp and c1 is the amended target filing's Accepted timestamp. All 936 public states match these timestamps, with $c0 < c1$ for every pair. In Figure 1, C(c) denotes the candidate filings for the routed issuer whose Accepted time is at or before c; structured facts additionally use explicit annual/quarterly form routing from the public question before ranking. Filing report-period metadata are recorded separately and do not provide a target-period hard filter for document routing. These cutoffs evaluate answering at specified filing states rather than discovery of an unknown revision time. SEC acceptance is the declared availability proxy: some submissions are disseminated on the next business day [22], as discussed in §6.1.

Evidence retrieval and recovery

Evidence-access conditions distinguish autonomous selection from reference-assisted selection. The autonomous conditions are cutoff BM25, dense retrieval with reranking, and the evidence-recovery baseline selected on development and then frozen. Recovery takes structured facts from the latest visible filing after public form routing

and filters them by the period, unit and any explicit entity scope parsed from the question. It retains dense retrieval with reranking after removing the hard lexical concept filter, and falls back to cutoff-BM25 passages when no eligible facts remain; its branch choice uses public evidence availability. The dense retriever is sentence-transformers/all-mpnet-base-v2; the reranker is cross-encoder/ms-marco-MiniLM-L-6-v2. Exact snapshots, inputs, and failure provenance are retained.

Recovery first obtains structured-fact candidates, filters and ranks them using the period, unit and explicit scope in the public question, and presents the leading facts as evidence cards. Cards retain value, unit, period and source-location fields; eligible facts supply the cards, while their absence triggers fallback to cutoff-BM25 passages. Evidence availability determines the branch without using answer correctness.

Reader and response records

The reader receives the public request and selected evidence and produces an answer or abstention under the fixed JSON schema. States are generated independently; the original-state prediction is not an input to amended-state answering. Exact messages, raw generations, model versions and input hashes are retained for offline numerical, unit, source and paired scoring. Reference-assisted conditions appear separately in the experiments, with decoding and evidence budgets fixed within each reader family.

S14.3 Evaluation details

The experiments use the resource to examine where answer gains occur and how errors change under different evidence access. Autonomous results, reference-assisted conditions and repeated development demonstrations retain separate identities.

Conditions and evaluation

Qwen2.5-7B-Instruct and Phi-3.5-mini-instruct use the same evidence and semantic instructions, with native chat templates, BF16 greedy decoding, at most five evidence items, an 8,192-token total budget and 512 reserved generation tokens [23, 24]. Main comparisons use a frozen protocol on the exposed ordinary test; Qwen reuses retained outputs. The 210 states form 105 pairs. All scheduled failures remain in their denominators. Detailed candidate pools, caching and costs are retained in Supplementary Section S13.

Autonomous conditions use cutoff BM25, dense retrieval with reranking, or frozen evidence recovery (§4). Reference-assisted conditions provide either the correct accession before BM25 searches within that document, or reference-located complete table/paragraph containers. The latter preserves headers and context without copying answer fields. Its five-container budget covers 200/210 test states and 134/144 development states; each remaining ten states is a retained failure. These are access conditions, not an additive causal decomposition of retrieval and reading losses.

Scoring distinguishes protocol compliance from the designated-state requirement. V checks output validity and citation/accession consistency with supplied evidence; N and U compare the finite-decimal value and canonical unit with the reference; A compares the returned accession with the state-designated accession. Primary numerical and unit scores are VN and VNU, and joint correctness is $J = V \times N \times U \times A$. Thus VN is a protocol-qualified numerical score. For Q0296, a response can give the shared correct value yet fail A by citing the original filing at the amended state. If its citation instead conflicts with its declared accession, it fails V earlier. Neither failure says that the number in the original filing is false.

Both-state correctness is $P = J(\text{original}) \times J(\text{amended})$, averaged over pairs. Four outcomes distinguish both correct, original only, amendment only and neither. Reference number/unit changes define the changed stratum. False changes require two valid, unit-comparable predictions on unchanged pairs; original-value reuse requires a valid, recognized-unit amended prediction matching the old reference but differing from the new one. Full and assessable denominators are retained. Resource-native-v1 supports USD, shares, CAD, USD/share and pure; the historical profile and parsed-field diagnostics remain in Supplementary Section S5/S7/S13.

Issuer-level paired analyses preserve both states and within-issuer dependence. Registered comparisons use issuer resampling and within-family Holm correction; seed SD describes training variation on fixed questions. The gain-heterogeneity analysis in Fig. 2 uses 20,000 joint issuer resamples and pointwise percentile intervals. Its

computation was frozen after the observed result directions were known. Full seeds, comparison families and inference limits are in Supplementary Section S10.2/S13.

Overall gains and paired errors

Evidence recovery improves joint and both-state correctness for both readers (Table 4), while the changed-answer stratum follows a different pattern.

Table 4. Answer components and both-state correctness under each evidence condition.

Note. V, VN, VNU and J use 210 scheduled states; P uses 105 pairs. V includes citation consistency. R indicates reference-assisted access, grouped separately within each reader. Resource-native-v1 is fixed across rows; its uniform rescore changes the historical Qwen complete-source score from 154 to 156 and P from 74 to 75, leaving autonomous scores unchanged.

Qwen recovery gains 42 correct states over BM25: +44 unchanged and -2 changed; Phi gains 52 unchanged and zero changed. On the 22 changed states, Qwen moves from 7 to 5 and Phi remains at 3. Figure 2 compares gain rates within strata rather than their unequal totals. The retrospective changed-minus-unchanged gain contrasts are -32.50 and -27.66 percentage points; the intervals describe this sample rather than new independent confirmation (Supplementary Section S10.2).

Pairing identifies errors beyond aggregate improvement. Recovery obtains 37/105 complete pairs for Qwen and 28/105 for Phi. It also produces false changes on 9/94 and 20/94 unchanged pairs, or 9/56 and 20/62 among unit-comparable valid pairs; original-reference reuse is 0/11 and 1/11. All six autonomous conditions retain the same strict ranking under state and paired scores. The additional information is the distribution of one-sided and shared failures, not a ranking reversal (Supplementary Table S14 and Fig. S1).

The failure partition also depends on scoring order. Across six conditions, 132 citation/accession inconsistencies enter the 623 protocol/precondition failures, whereas ten further failures reach the designated-accession check after matching number and unit. These are 1,260 condition-state records reusing 210 states. Supplementary Table S16 preserves the complete partition and parsed-field diagnostics, preventing the ten later failures from being read as all source errors.

Correct-filing and source-context access

Reference-assisted access improves both-state success on ordinary test: Qwen P increases from 13 with BM25 to 22 with the correct document and 75 with complete-source containers; Phi increases from 5 to 18 and 52 (Table 4). Under complete-source access, 32 of 188 valid Qwen outputs and 40 of 154 valid Phi outputs still fail the native number/unit requirement within all 210 states. These residual errors remain available for source-level inspection.

The development pattern differs. Complete-source Qwen/Phi scores are 67/144 and 35/144 versus recovery's 78/144 and 71/144; restricting to common coverage still gives 67 versus 74 and 35 versus 67 on 134 states. Development contains 38 mixed paragraph/table states, versus 12 in test. Source composition, length and reading behavior covary, so the contrasts identify an evidence-interface question rather than its cause. Complete containers are not guaranteed semantically sufficient. Detailed error counts and known-target retrieval coverage are retained in Supplementary Section S13; target coverage uses annotated locators, not pooled relevance judgments.

Adaptation and output controls on development

Development demonstrations examine whether changing evidence handling or output behavior improves the paired task. Rule-based single-accession binding raises Qwen joint correctness from 95 to 102/144 while changed-state correctness falls from 4 to 2/10. The corresponding LLM-parser increments are -1 and zero. Fixed one-epoch LoRA adaptations use 245 supported training states and 36 internal loss-holdout states [25]. Qwen's three seeds yield 111/112/112 and Phi's 106/108/106 on 144 development states, against controls of 102 and 90. Full results and selection scope are retained in Supplementary Section S1.

Contemporaneous free, JSON-envelope and numeric-string decoding controls separate output validity from joint correctness. Numeric constraints raise adapted Qwen validity from 130 to 144 but joint matches from 111 to 113; Phi validity rises from 140 to 144 and joint matches from 109 to 110. Changed-state counts stay fixed in all four

weight/model comparisons, and all four Holm-adjusted p-values are 1.00 with intervals including zero. These repeated-development controls establish neither equivalence nor a general improvement; their runtime is separate from earlier adaptation runs (Supplementary Section S3).

Source-linked cases

Source-linked cases make the distinction between a value error and a designated-source error inspectable.

In Q0691 under evidence recovery (Fig. 3), Qwen returns both reference answers, whereas Phi cites the amended accession with the original value. In Q0296, the two covers share USD 197,795,766, but Qwen BM25 cites the original accession at the amended state. Cases were selected by fixed ID hashes within observed outcome categories. Official URLs, cutoffs, hashes, table locators and predictions remain available; the figure is a verified source-cell transcription, not a SEC screenshot (Supplementary Section S5.3).

S14.4 Conclusion and scope

FilingRevisionQA links original and amended financial filings through traceable question pairs and executable evaluation of numerical answers, units, designated sources and both-state correctness. Across the evaluated readers, positive net recovery gains over cutoff-filtered BM25 arise from unchanged-answer states, while the joint-correct count on the 22 changed test states declines or stays constant. Researchers can use paired targets and controlled evidence access to compare retrieval and reading components and trace failures to their source filings.

Scope and limitations

The evidence supports reuse of the paired resource and diagnosis of the evaluated systems within the following scope. Existing temporal, traceable-source, retrieval, and adaptation work supplies the foundations; the contribution here is the paired resource, executable evaluation, and observed failure structure.

Population and change coverage. The collection comprises selected traceable SEC pairs, rather than a representative sample of all disclosures. Changed-answer analysis uses five development and eleven test pairs. Selection and support requirements, sparse changes, and unresolved event categories limit population estimates.

Review provenance. Records and author attestations support the review account. The projected-wording review is supported by the retained batch-level author confirmation; its evidence granularity is documented in Supplementary Section S5.2. AI screening, semantic explanations, and source checks remain distinct from human judgments.

Evaluation history. Development was repeatedly reused, and later ordinary-test comparisons are supplementary. Frozen protocols preserve experiment identity. Multiplicity corrections apply within declared comparison families; historical selection dependence remains. The inspected internal holdout, whose supported subset was used for final teacher-forced loss evaluation, is a diagnostic partition. Optimizer seeds measure training variability on fixed questions; two reader families leave wider model generalization open.

Temporal eligibility. Accepted timestamps operationalize eligibility; zero clock violations establish compliance with this declared proxy and frozen corpus, while exact historical public availability remains unverified across all states.

Source verification and evidence conditions. Structural source verification establishes parsing and localization. Complete containers may still lack sufficient semantic context. Their scores describe the supplied evidence condition rather than a universal performance ceiling, and the covarying source composition, length, question, and reader behavior preclude an additive causal error budget.

Adaptation and interventions. Adaptation findings are conditional on the training subset selected by the source-support rule and on the chosen evidence interface. The copying control, incomplete citation support, a single identifier permutation, coupled masking changes, and specific decoder constraints limit reasoning and causal interpretations. Pretraining overlap remains possible. Model families differ in tokenizer and parameter size; the adaptation runs also used different hardware.

Responsible use. The resource is intended for research. Amendment status alone establishes none of fraud, misconduct, audit failure, or investment risk.

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